Supplemental Material for "Absorbing phase transition in a queueing model of coupled adaptive agents"

Alexei Vazquez[1,*]

[1] *Nodes & Links Ltd, Salisbury House, Station Road, Cambridge, CB1 2LA, UK*

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The four tables below are reference material for the main text, which quotes only the entries its argument uses and reports them here in full. The first two are the parameter scans; the last two are the criterion-robustness and finite-size scans behind Sec. VI A. Every entry is produced from the stored simulation output by `make_supplementary.py` in the code repository, `https://github.com/av2atgh/interagents`; nothing is transcribed by hand. Section, equation and table numbers with no "Supplementary" prefix refer to the main text.

* alexei@nodeslinks.com

Supplementary Table I. Sensitivity of the pair at the reference point ($L = 2$, $c = 0.35$, $R_I = 0.8428$, $\epsilon = 10^{-2}$, $\tau_{\mathrm{mem}} = 200$), one parameter at a time, N independent realisations each. "collapsed" is the fraction of realisations that reached the solitary state within the run; $\langle T_c \rangle$ is the censored-exponential mean collapse time in steps, with a lower bound given where no collapse was observed; ϕ is the occupancy, with the standard error in units of the last digit. Runs are 2×10^5 steps except in the last block, where the run length is the variable. Note that $\langle T_c \rangle$ is independent of the run length while ϕ is not.

	value	N	collapsed	$\langle T_c \rangle$	ϕ
τ_{mem}	25	250	1.000	$9.25(\pm0.58) \times 10^2$	0.024(1)
	50	250	1.000	$4.74(\pm0.30) \times 10^3$	0.043(1)
	100	250	0.996	$3.20(\pm0.20) \times 10^4$	0.175(10)
	200	250	0.200	$8.99(\pm1.27) \times 10^5$	0.899(15)
	400	250	0.000	$> 5 \times 10^7$	1.000(1)
	800	250	0.000	$> 5 \times 10^7$	1.000(1)
R_I	0.8400	250	0.612	$2.18(\pm0.18) \times 10^5$	0.664(21)
	0.8428	250	0.200	$8.99(\pm1.27) \times 10^5$	0.899(15)
	0.8500	250	0.004	$4.99(\pm4.99) \times 10^7$	0.997(3)
	0.8700	250	0.000	$> 5 \times 10^7$	1.000(1)
	0.9000	250	0.000	$> 5 \times 10^7$	1.000(1)
	0.9500	250	0.000	$> 5 \times 10^7$	1.000(1)
ϵ	0	250	0.768	$1.41(\pm0.10) \times 10^5$	0.541(21)
	0.001	250	0.740	$1.55(\pm0.11) \times 10^5$	0.576(22)
	0.01	250	0.200	$8.99(\pm1.27) \times 10^5$	0.899(15)
	0.03	250	0.000	$> 5 \times 10^7$	1.000(1)
	0.1	250	0.000	$> 5 \times 10^7$	1.000(1)
$\hat{p}(0)$	0.10	250	1.000	$2.50(\pm0.16) \times 10^3$	0.020(1)
	0.30	250	1.000	$2.50(\pm0.16) \times 10^3$	0.020(1)
	0.50	250	0.720	$9.06(\pm0.68) \times 10^4$	0.329(28)
	0.70	250	0.200	$8.99(\pm1.27) \times 10^5$	0.899(15)
	0.85	250	0.200	$8.99(\pm1.27) \times 10^5$	0.899(15)
	1.00	250	0.200	$8.99(\pm1.27) \times 10^5$	0.899(15)
T	2.5×10^4	250	0.032	$7.74(\pm2.74) \times 10^5$	0.988(4)
	5×10^4	250	0.048	$1.02(\pm0.29) \times 10^6$	0.974(8)
	10^5	250	0.100	$9.49(\pm1.90) \times 10^5$	0.948(11)
	2×10^5	250	0.200	$8.99(\pm1.27) \times 10^5$	0.899(15)
	4×10^5	250	0.396	$8.03(\pm0.81) \times 10^5$	0.797(19)
	8×10^5	250	0.620	$8.31(\pm0.67) \times 10^5$	0.649(22)

Supplementary Table II. Network results against run length, initial belief and the remaining parameters. Entries are the coupled edge fraction with the standard error in units of the last digit, on the randomized network at $n = 600$, $\langle k \rangle = 4$, over 8 to 24 independent realisations per entry. Unless it is the variable, the run is 4×10^5 steps, the initial belief 0.85, $\epsilon = 10^{-2}$, $\tau_{\mathrm{mem}} = 200$, $R_I = 0.95$ and $L = 2$. A dash means the combination was not run.

run length T (steps)								
c	2.5×10^4	5×10^4	10^5	2×10^5	4×10^5	8×10^5	1.6×10^6	6.4×10^6
0.002	–	–	–	–	0.992(1)	–	–	–
0.010	–	–	–	–	0.960(2)	–	–	–
0.015	0.857(3)	0.883(3)	0.897(2)	0.904(3)	0.904(2)	0.910(3)	0.909(3)	–
0.020	0.593(2)	0.672(2)	0.756(3)	0.816(3)	0.849(2)	0.866(3)	0.878(3)	–
0.025	0.283(2)	0.295(2)	0.346(2)	0.424(4)	0.522(2)	0.628(4)	0.732(3)	0.841(4)
0.030	0.178(2)	0.155(2)	0.132(2)	0.114(2)	0.122(2)	0.174(2)	0.249(3)	0.415(5)
0.035	–	–	–	–	–	–	0.004(1)	0.003(1)
0.045	–	–	–	–	0.000(1)	–	0.000(1)	0.000(1)
0.062	–	–	–	–	–	–	0.000(1)	0.000(1)
0.070	–	–	–	–	0.000(1)	–	–	–

initial belief $\hat{p}(0)$				
c	0.1	0.3	0.5	0.85
0.002	–	–	–	0.992(1)
0.010	0.963(3)	0.961(2)	0.961(3)	0.960(2)
0.015	0.907(3)	0.907(2)	0.908(3)	0.904(2)
0.020	0.848(3)	0.851(4)	0.852(3)	0.849(2)
0.025	0.524(3)	0.520(4)	0.522(4)	0.522(2)
0.030	0.118(2)	0.119(3)	0.116(3)	0.122(2)
0.045	–	–	–	0.000(1)
0.070	–	–	–	0.000(1)

ϵ					
c	0	0.003	0.01	0.03	0.1
0.002	–	–	0.992(1)	–	–
0.010	0.000(1)	0.898(4)	0.960(2)	0.996(2)	0.987(3)
0.015	0.000(1)	0.339(5)	0.904(2)	0.983(3)	0.986(2)
0.020	0.000(1)	0.003(1)	0.849(2)	0.961(4)	0.986(3)
0.025	0.000(1)	0.000(1)	0.522(2)	0.927(5)	0.985(3)
0.030	0.000(1)	0.000(1)	0.122(2)	0.889(5)	0.985(3)
0.045	–	–	0.000(1)	–	–
0.070	–	–	0.000(1)	–	–

τ_{mem}					
c	50	100	200	400	800
0.002	–	–	0.992(1)	–	–
0.010	0.996(1)	0.984(3)	0.960(2)	0.946(3)	0.944(3)
0.015	0.989(2)	0.963(3)	0.904(2)	0.835(5)	0.746(3)
0.020	0.979(3)	0.939(4)	0.849(2)	0.481(2)	0.570(3)
0.025	0.966(4)	0.910(5)	0.522(2)	0.267(1)	0.450(2)
0.030	0.951(4)	0.821(5)	0.122(2)	0.212(2)	0.341(1)
0.045	–	–	0.000(1)	–	–
0.070	–	–	0.000(1)	–	–

R_I				
c	0.85	0.9	0.95	1
0.002	–	–	0.992(1)	–
0.010	0.934(4)	0.950(4)	0.960(2)	0.972(3)
0.015	0.863(4)	0.887(4)	0.904(2)	0.925(4)
0.020	0.704(4)	0.801(5)	0.849(2)	0.876(4)
0.025	0.233(1)	0.379(4)	0.522(2)	0.662(5)
0.030	0.004(1)	0.031(2)	0.122(2)	0.243(3)
0.045	–	–	0.000(1)	–
0.070	–	–	0.000(1)	–

L			
c	2	3	4
0.002	0.992(1)	–	–
0.010	0.960(2)	0.385(3)	0.000(1)
0.015	0.904(2)	0.000(1)	0.000(1)
0.020	0.849(2)	0.000(1)	0.000(1)
0.025	0.522(2)	0.000(1)	0.000(1)
0.030	0.122(2)	0.000(1)	0.000(1)
0.045	0.000(1)	–	–
0.070	0.000(1)	–	–

Supplementary Table III. Robustness of the coupled-edge criterion. Giant component S of the coupled subgraph on the randomized network, $n = 600$, 4×10^5 steps, $N = 24$ realisations, as the rate w above which an edge is called coupled is varied across the gap between the two bands of Fig. 5(b) of the main text. The value used throughout is $10\epsilon^2 = 10^{-3}$. Over the decade from 3×10^{-4} to 3×10^{-3} the order parameter moves by less than 0.012 in the coupled phase; outside it the criterion fails, the exploration floor percolating by itself at a cut of ϵ^2 and genuinely coupled edges being deleted at 5×10^{-3}.

c	10^{-4}	3×10^{-4}	10^{-3}	2×10^{-3}	3×10^{-3}	5×10^{-3}
0.002	1.000(1)	1.000(1)	1.000(1)	1.000(1)	0.999(1)	0.963(3)
0.010	1.000(1)	1.000(1)	1.000(1)	1.000(1)	0.998(1)	0.947(3)
0.015	1.000(1)	1.000(1)	1.000(1)	0.999(1)	0.997(1)	0.955(3)
0.020	1.000(1)	1.000(1)	1.000(1)	0.997(1)	0.988(2)	0.849(8)
0.025	0.987(1)	0.932(3)	0.831(5)	0.662(9)	0.399(26)	0.049(4)
0.030	0.685(7)	0.011(1)	0.009(1)	0.008(1)	0.007(1)	0.006(1)
0.045	0.494(12)	0.000(1)	0.000(1)	0.000(1)	0.000(1)	0.000(1)
0.070	0.499(13)	0.000(1)	0.000(1)	0.000(1)	0.000(1)	0.000(1)

Supplementary Table IV. Finite size. Giant component S of the coupled subgraph at $\langle k \rangle = 4$, $R_I = 0.95$, 4×10^5 steps, with the standard error in units of the last digit, over $N = 24$ realisations at $n = 300$ and 600, 16 at 1200, 12 at 2400 and 8 at 4800. The randomized network shows no size dependence; under triadic closure S falls systematically with n at every cost below the transition. Figure 6 of the main text plots the same measurement over the full range of c.

	c	300	600	1200	2400	4800
$LS(\ell = 1)$	0.015	0.953(18)	0.843(32)	0.835(38)	0.681(58)	0.594(86)
	0.020	0.936(24)	0.882(22)	0.705(41)	0.601(51)	0.370(77)
	0.025	0.305(22)	0.227(20)	0.143(14)	0.091(15)	0.049(8)
	0.030	0.017(1)	0.010(1)	0.006(1)	0.003(1)	0.002(1)
randomized	0.015	1.000(1)	1.000(1)	1.000(1)	1.000(1)	1.000(1)
	0.020	1.000(1)	1.000(1)	1.000(1)	1.000(1)	1.000(1)
	0.025	0.823(9)	0.831(5)	0.825(6)	0.820(3)	0.823(5)
	0.030	0.019(1)	0.009(1)	0.005(1)	0.003(1)	0.002(1)